

MATH141 Final Exam Study Guide

Detailed 3-day crash course with mathematical symbols, not plain-text formula notation

This guide explains the included portion as if you did not attend class. It covers Tutorials 4–8, Quiz 2, the final exam practice questions, and the formula booklet. Every question is explained by method, steps, answer, and a common exam warning where useful.

I use mathematical symbols such as $\int$, $\sqrt{}$, π , θ , 2 , 3 , $\frac{1}{2}$, $\leq$, $\parallel$, $\cdot$, $\times$, and vector notation. Formula booklet references are given by page and visible section heading.

3-day study plan

- Day 1: Integration, substitution, trig integrals, area, distance from velocity, and the Fundamental Theorem of Calculus. Redo Tutorial 4, Quiz 2 Q1–Q2, and Practice Q1–Q3.
- Day 2: Complex numbers, complex square roots/quadratics, polar form, spheres, vectors, dot product, cross product, projections, and vector applications. Redo Tutorial 5–6, Quiz 2 Q3–Q6, and Practice Q4–Q9.
- Day 3: Lines, planes, distances, Gaussian elimination, matrices, determinants, and inverse existence. Redo Tutorial 7–8 and Practice Q10–Q15.

Common mistake: Reading is not enough. Redo each calculation on paper and write the method: substitution, limits, normal vector, direction vector, row operations, or determinant expansion.

Formula booklet map

Page 1 — Pythagorean identities	$\sin^2\theta + \cos^2\theta = 1$; $1 + \tan^2\theta = \sec^2\theta$; $1 + \cot^2\theta = \csc^2\theta$.
Page 1 — Double-angle identities	$\sin 2\theta = 2\sin\theta\cos\theta$; $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$; $\tan 2\theta = 2\tan\theta/(1 - \tan^2\theta)$.
Page 1 — Trig derivatives	$(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$.
Page 1 — Basic integration	$\int x^n dx = x^{n+1}/(n+1) + C$ for $n \neq -1$; $\int 1/x dx = \ln x + C$; $\int \sin x dx = -\cos x + C$; $\int \cos x dx = \sin x + C$; $\int \tan x dx = -\ln \cos x + C$.
Page 1 — FTC	$d/dx \int \text{from } u(x) \text{ to } v(x) \text{ of } f(t) dt = v'(x)f(v(x)) - u'(x)f(u(x))$.
Page 2 — Complex numbers	$z = a + bi$, $\bar{z} = a - bi$, $ z = \sqrt{a^2 + b^2}$, polar form $z = r(\cos\theta + i \sin\theta)$.
Page 2 — 3D space	Sphere: $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$.
Page 2 — Dot product	$a \cdot b = \ a\ \ b\ \cos\theta$. Perpendicular means $a \cdot b = 0$.
Page 2 — Cross product	$a \times b$ is perpendicular to a and b . Triangle area = $\frac{1}{2} \ PQ \times PR\ $.
Page 3 — Lines, planes, distances	Line $r = r_0 + tv$. Plane $n \cdot (r - r_0) = 0$. Point-plane distance = $ Ax_0 + By_0 + Cz_0 + D /\sqrt{A^2 + B^2 + C^2}$.

Core lessons you need before solving

Integration

Integration reverses differentiation. The power rule works after rewriting radicals and fractions as powers. A definite integral $\int_a^b f(x) dx$ means accumulated signed area: find an antiderivative F , then compute $F(b) - F(a)$.

$$\sqrt{x} = x^{1/2}, \quad 1/x^3 = x^{-3}, \quad \sqrt[3]{x^2} = x^{2/3}$$

Formula sheet: Page 1 → Basic integration and FTC.

Substitution

Substitution is reverse chain rule. Choose u as the inside function when its derivative also appears. Replace dx correctly. For definite integrals, change the bounds immediately.

$$u = x^4 + 1 \Rightarrow du = 4x^3 dx, \text{ so } x^3 dx = du/4$$

Trig integrals

For odd powers of $\sin$ or $\cos$, save one factor for du and use $\sin^2 x + \cos^2 x = 1$. For $\tan/\sec$ integrals, save $\sec^2 x dx$ for $u = \tan x$, or save $\sec x \tan x dx$ for $u = \sec x$. For even powers, double-angle identities usually help.

Formula sheet: Page 1 → Pythagorean identities, double-angle identities, trig derivatives.

Complex numbers

Write every complex expression as real part plus imaginary part. If $A+Bi=C+Di$, then $A=C$ and $B=D$. For square roots, set $\sqrt{(a+bi)}=x+yi$ and square both sides.

$$(x+yi)(2+i)=(2x-y)+i(x+2y)$$

Formula sheet: [Page 2 → Complex numbers.](#)

Vectors, lines, and planes

A vector from P to Q is $\mathbf{Q}-\mathbf{P}$. Dot product gives angle and perpendicular tests. Cross product gives a perpendicular vector, plane normals, and area. A line needs a point and direction. A plane needs a point and normal.

$$\mathbf{PQ}=\mathbf{Q}-\mathbf{P}, \mathbf{a} \cdot \mathbf{b}=\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta, \text{ area of triangle}=\frac{1}{2}\|\mathbf{PQ} \times \mathbf{PR}\|$$

Formula sheet: [Page 2 → Dot product and Cross product.](#) [Page 3 → Lines and Planes.](#)

Linear algebra

Row reduction is organized elimination. Pivot variables are solved; free variables become parameters. A square matrix has an inverse exactly when its determinant is nonzero.

$$\mathbf{A}^{-1} \text{ does not exist} \Leftrightarrow \det(\mathbf{A})=0$$

Tutorial 4 — Integration

T4 Q1a

Question: Evaluate $\int (x^2-4)^{2/3} \sqrt{x^2} \, dx$.

Big idea: Rewrite the denominator as a power and split the fraction.

Explanation

- $\sqrt[3]{x^2}=x^{2/3}$, so divide both numerator terms by $x^{2/3}$.

$$\int (x^{4/3}-4x^{-2/3})dx = 3/7x^{7/3}-12x^{1/3}+C$$

Answer: $3/7x^{7/3}-12x^{1/3}+C$

Formula sheet: [Page 1 → Basic integration.](#)

Common mistake: Do not cancel x^2-4 with the radical; split term by term.

T4 Q1b

Question: Evaluate $\int_{-3}^0 (1+\sqrt{9-x^2})dx$.

Big idea: Split the integral and recognize a quarter-circle.

Explanation

- $\int_{-3}^0 1dx=3$. The graph $y=\sqrt{9-x^2}$ is the upper half of $x^2+y^2=9$, a circle of radius 3.
- The interval -3 to 0 covers one quarter of that circle.

$$\text{area} = 3 + \frac{1}{4}\pi(3)^2 = 3 + 9\pi/4$$

Answer: $3+9\pi/4$

Formula sheet: [Page 1 → FTC; circle area is geometry.](#)

Common mistake: Do not use the full semicircle; it is only a quarter circle.

T4 Q1c

Question: Evaluate $\int x^2 \sqrt{x+1} dx$.

Big idea: Substitute $u=x+1$ and replace x by $u-1$.

Explanation

- Let $u=x+1$, so $x=u-1$ and $dx=du$.

$$\int (u-1)^2 \sqrt{u} \, du = \int (u^{3/2}-2u^{1/2}+u^{-1/2})du$$

- Integrate powers, then substitute $u=x+1$.

Answer: $2/5(x+1)^{5/2}-4/3(x+1)^{3/2}+2(x+1)^{1/2}+C$

Formula sheet: [Page 1 → Basic integration.](#)

Common mistake: After $u=x+1$, x^2 becomes $(u-1)^2$.

T4 Q1d

Question: Evaluate $\int \sin^3 x \cos x \, dx$.

Big idea: Use $\sin^2 x = 1 - \cos^2 x$ and $u = \cos x$.

Explanation

- Write $\sin^3 x = (1 - \cos^2 x)\sin x$. Let $u = \cos x$, $du = -\sin x \, dx$.

$$\int \sin^3 x \cos x \, dx = \int (u - 1/u) du = \frac{1}{2}u^2 - \ln|u| + C$$

Answer: $\frac{1}{2}\cos^2 x - \ln|\cos x| + C$

Formula sheet: Page 1 → Pythagorean identities and log integration.

Common mistake: Track the negative sign from $du = -\sin x \, dx$.

T4 Q1e

Question: Evaluate $\int x^{23} \sqrt{x^3 + 5} \, dx$.

Big idea: The derivative of $x^3 + 5$ is $3x^2$.

Explanation

- Let $u = x^3 + 5$, so $du = 3x^2 \, dx$ and $x^2 \, dx = du/3$.

$$\frac{1}{3} \int u^{1/3} \, du = \frac{1}{4} u^{4/3} + C$$

Answer: $\frac{1}{4}(x^3 + 5)^{4/3} + C$

Formula sheet: Page 1 → Basic integration.

Common mistake: Do not lose the factor $\frac{1}{3}$.

T4 Q2

Question: Work: $W = \int_0^{36} (300\sqrt{x} - 10x\sqrt{x}) \, dx$.

Big idea: Work is accumulated force, so integrate force over distance.

Explanation

- Rewrite $\sqrt{x} = x^{1/2}$ and $x\sqrt{x} = x^{3/2}$.

$$W = [200x^{3/2} - 4x^{5/2}]_0^{36}$$

- Because $36^{3/2} = 216$ and $36^{5/2} = 7776$.

Answer: 12096

Formula sheet: Page 1 → Basic integration.

Common mistake: Plug in bounds after finding the antiderivative.

T4 Q3

Question: Area under $y = \sqrt{25 - x^2}$ from $x = 0$ to $x = 5$.

Big idea: This curve is a quarter of a circle of radius 5.

Explanation

- $x^2 + y^2 = 25$ is a circle. Since $y \geq 0$ and $0 \leq x \leq 5$, the region is one quarter of the circle.

$$\text{area} = \frac{1}{4}\pi(5)^2 = 25\pi/4$$

Answer: $25\pi/4$

Formula sheet: Page 1 → definite integrals; geometry gives the area.

Common mistake: Do not take a half-circle area.

T4 Q4

Question: $v(t) = t - t^2$ on $0 \leq t \leq 2$. Find distance travelled.

Big idea: Distance uses speed $|v(t)|$, so split where velocity changes sign.

Explanation

- $v(t) = t(1 - t)$, so $v \geq 0$ on $[0, 1]$ and $v \leq 0$ on $[1, 2]$.

$$\text{distance} = \int_0^1 (t - t^2) \, dt + \int_1^2 (t^2 - t) \, dt = \frac{1}{6} + \frac{5}{6}$$

Answer: 1

Formula sheet: Page 1 → Basic integration and FTC.

Common mistake: $\int$ velocity gives displacement, not distance.

T4 Q5a

Question: Differentiate $\int$ from $1/x$ to $1/\sqrt{x}$ of $\sin(t^2) \, dt$.

Big idea: FTC with moving upper and lower bounds.

Explanation

- Upper $v=x^{-1/2}$, lower $u=x^{-1}$.

$$F'(x)=v'(x)\sin(v^2)-u'(x)\sin(u^2)$$

- $v'=-1/(2x^{3/2})$, $u'=-1/x^2$.

Answer: $-\sin(1/x)/(2x^{3/2})+\sin(1/x^2)/x^2$

Formula sheet: Page 1 → FTC.

Common mistake: The lower-limit term is subtracted.

T4 Q5b

Question: $g(x)=\int$ from $\cos x$ to $\sin x$ of $\ln(2+3t)dt$. Find $g'(0)$.

Big idea: FTC plus derivatives of sin and cos.

Explanation

$$g'(x)=\cos x \cdot \ln(2+3\sin x) - (-\sin x) \cdot \ln(2+3\cos x)$$

- At $x=0$, $\cos 0=1$ and $\sin 0=0$.

Answer: $\ln 2$

Formula sheet: Page 1 → FTC and trig derivatives.

Common mistake: Do not integrate $\ln(2+3t)$; differentiate using FTC.

T4 Q6a

Question: Evaluate $\int \sqrt{\sin x} \cos^5 x \, dx$.

Big idea: Save one $\cos x \, dx$ and convert $\cos^4 x$.

Explanation

- $\cos^5 x = \cos^4 x \cos x = (1-\sin^2 x)^2 \cos x$. Let $u=\sin x$.

$$\int u^{1/2}(1-u^2)^2 du = \int (u^{1/2}-2u^{5/2}+u^{9/2}) du$$

Answer: $\frac{2}{3}\sin^{3/2}x - \frac{4}{7}\sin^{7/2}x + \frac{2}{11}\sin^{11/2}x + C$

Formula sheet: Page 1 → Pythagorean identities.

Common mistake: Save one $\cos x$ for du .

T4 Q6b

Question: Evaluate $\int \sin^2 x \cos^2 x \, dx$.

Big idea: Use double-angle identities for even powers.

Explanation

- $\sin^2 x \cos^2 x = \frac{1}{4}\sin^2 2x$ and $\sin^2 2x = (1-\cos 4x)/2$.

$$\int (1-\cos 4x)/8 \, dx = x/8 - \sin 4x/32 + C$$

Answer: $x/8 - \sin 4x/32 + C$

Formula sheet: Page 1 → Double-angle identities.

Common mistake: Include the chain-rule divisor 4 when integrating $\cos 4x$.

T4 Q6c

Question: Evaluate $\int \sin 7x \cos 3x \, dx$.

Big idea: Use product-to-sum.

Explanation

- $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$.

$$\int \sin 7x \cos 3x \, dx = \frac{1}{2} \int (\sin 10x + \sin 4x) dx$$

Answer: $-\cos 10x/20 - \cos 4x/8 + C$

Formula sheet: Page 1 → trig identities and integration.

Common mistake: Product-to-sum is not printed, but it follows from trig identities.

Tutorial 5 — Trig integration and complex numbers

T5 Q1a

Question: Evaluate $\int \tan^2 x \sec^4 x \, dx$.

Big idea: Save $\sec^2 x \, dx$ and set $u = \tan x$.

Explanation

- $\sec^4 x = \sec^2 x (1 + \tan^2 x)$.

$$\int u^2(1+u^2)du = \frac{1}{3}u^3 + \frac{1}{5}u^5 + C$$

Answer: $\frac{1}{3}\tan^3 x + \frac{1}{5}\tan^5 x + C$

Formula sheet: Page 1 $\rightarrow 1 + \tan^2 \theta = \sec^2 \theta$; derivative of $\tan$.

Common mistake: Leave one $\sec^2 x \, dx$ for du .

T5 Q1b

Question: Evaluate $\int \tan^3 x \sec^3 x \, dx$.

Big idea: Save $\sec x \tan x \, dx$ and set $u = \sec x$.

Explanation

- $\tan^3 x \sec^3 x = \tan^2 x \sec^2 x (\sec x \tan x)$. Use $\tan^2 x = \sec^2 x - 1$.

$$\int (u^2 - 1)u^2 du = \frac{1}{5}u^5 - \frac{1}{3}u^3 + C$$

Answer: $\frac{1}{5}\sec^5 x - \frac{1}{3}\sec^3 x + C$

Formula sheet: Page 1 $\rightarrow$ Pythagorean identity.

Common mistake: $u = \tan x$ leaves an awkward $\sec x$; $u = \sec x$ is cleaner.

T5 Q2a

Question: Find x, y : $x(1+i)^2 + y(2-i)^2 = 3+10i$.

Big idea: Expand and match real/imaginary parts.

Explanation

- $(1+i)^2 = 2i$ and $(2-i)^2 = 3-4i$.

$$3y + i(2x - 4y) = 3 + 10i$$

- So $3y = 3$ and $2x - 4y = 10$.

Answer: $x = 7, y = 1$

Formula sheet: Page 2 $\rightarrow$ Complex numbers.

Common mistake: Use $i^2 = -1$.

T5 Q2b

Question: Find x, y : $1/(x+iy) + 1/(1+2i) = 1$.

Big idea: Rationalize the known complex fraction.

Explanation

- $1/(1+2i) = (1-2i)/5$.

$$1/(x+iy) = 1 - (1-2i)/5 = 4/5 + 2i/5$$

- Invert to get $x+iy = 1-i/2$.

Answer: $x = 1, y = -\frac{1}{2}$

Formula sheet: Page 2 $\rightarrow$ Complex numbers.

Common mistake: Multiply by conjugates when dividing complex numbers.

T5 Q2c

Question: Solve $z - 8 = i(7 - 2\bar{z})$ and find $|z|$.

Big idea: Use $z = x+iy$ and $\bar{z} = x-iy$.

Explanation

- Right side $i(7 - 2x + 2iy) = -2y + i(7 - 2x)$.

- Match $x - 8 = -2y$ and $y = 7 - 2x$.

Answer: $z = 2 + 3i, |z| = \sqrt{13}$

Formula sheet: Page 2 $\rightarrow$ Complex numbers.

Common mistake: Conjugate changes the sign of the imaginary part.

T5 Q3a

Question: Solve $2z^2 - 2iz - 5 = 0$.

Big idea: Use the quadratic formula.

Explanation

$$z = [2i \pm \sqrt{(-2i)^2 - 4(2)(-5)})]/4$$

- The discriminant is 36.

Answer: $z = 3/2 + i/2$ or $z = -3/2 + i/2$

Formula sheet: Page 2 → Complex numbers.

Common mistake: $(-2i)^2 = -4$, not 4.

T5 Q3b

Question: Solve $z^2 - (1 - 4i)z - (5 - i) = 0$.

Big idea: Complex quadratic simplification.

Explanation

- Apply the quadratic formula and simplify, or check by substitution.
- The two roots make the polynomial zero.

Answer: $z = -1 - i$ or $z = 2 - 3i$

Formula sheet: Page 2 → Complex numbers.

Common mistake: Combine real and imaginary parts carefully.

T5 Q3c

Question: Solve $z^2 - z + 8 + 2(z + 1)i = 0$.

Big idea: Collect z terms first.

Explanation

- Expand $2(z + 1)i = 2iz + 2i$.

$$z^2 + (-1 + 2i)z + (8 + 2i) = 0$$

Answer: $z = 2i$ or $z = 1 - 4i$

Formula sheet: Page 2 → Complex numbers.

Common mistake: Do not apply the quadratic formula before collecting terms.

T5 Q4a

Question: Write $1 - i$ in polar form.

Big idea: Find modulus and quadrant angle.

Explanation

- $r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$. The point is in quadrant IV, so $\theta = 7\pi/4$ or $\theta = -\pi/4$.

Answer: $\sqrt{2}(\cos(7\pi/4) + i \sin(7\pi/4))$

Formula sheet: Page 2 → Complex numbers.

Common mistake: Use the correct quadrant.

T5 Q4b

Question: Write $-2 + 2\sqrt{3}i$ in polar form.

Big idea: Find modulus and angle in quadrant II.

Explanation

- $r = \sqrt{4 + 12} = 4$. The reference angle is $\pi/3$ and the quadrant II angle is $2\pi/3$.

Answer: $4(\cos(2\pi/3) + i \sin(2\pi/3))$

Formula sheet: Page 2 → Complex numbers.

Common mistake: A negative real part and positive imaginary part means quadrant II.

Tutorial 6 — Vectors and 3D space

T6 Q1a

Question: Sphere $x^2+y^2+z^2-4x-6z-3=0$.

Big idea: Complete squares.

Explanation

- $x^2-4x=(x-2)^2-4$ and $z^2-6z=(z-3)^2-9$.

$$(x-2)^2+y^2+(z-3)^2=16$$

Answer: center (2,0,3), radius 4

Formula sheet: Page 2 → 3D space.

Common mistake: Radius is $\sqrt{16}$, not 16.

T6 Q1b

Question: Sphere $x^2+y^2+z^2+2y+4z=20$.

Big idea: Complete squares.

Explanation

- $y^2+2y=(y+1)^2-1$ and $z^2+4z=(z+2)^2-4$.

$$x^2+(y+1)^2+(z+2)^2=25$$

Answer: center (0, -1, -2), radius 5

Formula sheet: Page 2 → 3D space.

Common mistake: Inside signs are opposite the center coordinates.

T6 Q2

Question: Sphere with diameter A(2,1,4), B(4,3,10).

Big idea: Midpoint is center, half distance is radius.

Explanation

- $C=(3,2,7)$. $AB=\sqrt{(2^2+2^2+6^2)}=2\sqrt{11}$, so $r=\sqrt{11}$.

Answer: $(x-3)^2+(y-2)^2+(z-7)^2=11$

Formula sheet: Page 2 → 3D space.

Common mistake: Right side is r^2 .

T6 Q3

Question: P(1, -1, 0), Q(2, 1, -1), R(-1, 1, 3).

Big idea: Subtract points, then use dot/cross products.

Explanation

- $PQ=\langle 1, 2, -1 \rangle$ and $PR=\langle -2, 2, 3 \rangle$.
- $PQ \cdot PR = -1$. $PQ \times PR = \langle 8, -1, 6 \rangle$.

$$\text{area} = \frac{1}{2}\sqrt{(8^2+(-1)^2+6^2)} = \sqrt{101}/2$$

Answer: $PQ=\langle 1, 2, -1 \rangle$, $PR=\langle -2, 2, 3 \rangle$, $\text{dot}=-1$, $\text{area}=\sqrt{101}/2$

Formula sheet: Page 2 → Dot product and Cross product.

Common mistake: Use vectors from the same starting point.

T6 Q4a

Question: $\|u\|=1/2$, $\|v\|=2$, $\theta=\pi/4$. Find $\|u-v\|$.

Big idea: Expand squared magnitude.

Explanation

$$\|u-v\|^2 = \|u\|^2 - 2\|u\|\|v\|\cos\theta + \|v\|^2$$

$$\frac{1}{4} - 2(\frac{1}{2})(2)\cos(\pi/4) + 4 = 17/4 - \sqrt{2}$$

Answer: $\sqrt{(17/4 - \sqrt{2})}$

Formula sheet: Page 2 → Dot product.

Common mistake: The middle term is negative for $u-v$.

T6 Q4b

Question: Unit u, v with $\theta=2\pi/3$. Find $\|u+2v\|$.

Big idea: Expand dot product.

Explanation

$$\|u+2v\|^2=1+4(u\cdot v)+4$$

- $u\cdot v=\cos(2\pi/3)=-\frac{1}{2}$, so the square is 3.

Answer: $\sqrt{3}$

Formula sheet: Page 2 → Dot product.

Common mistake: The coefficient 2 gives $4u\cdot v$ and $4\|v\|^2$.

T6 Q4c

Question: Angle between $a=(2,2,-1)$ and $b=(1,2,1)$.

Big idea: Use $\cos\theta=(a\cdot b)/(\|a\|\|b\|)$.

Explanation

- $a\cdot b=5$, $\|a\|=3$, $\|b\|=\sqrt{6}$.

$$\cos\theta=5/(3\sqrt{6})$$

Answer: $\theta=\arccos(5/(3\sqrt{6}))$

Formula sheet: Page 2 → Dot product.

Common mistake: The angle is arccos of the ratio.

T6 Q4d

Question: Find x so $\langle 2x, x+8, 7 \rangle$ and $\langle -3, 2, x \rangle$ are perpendicular.

Big idea: Perpendicular means dot product zero.

Explanation

$$-6x+2(x+8)+7x=3x+16=0$$

Answer: $x=-16/3$

Formula sheet: Page 2 → Dot product.

Common mistake: Distribute $2(x+8)$.

T6 Q4e

Question: $a=(5,-2)$ and $b=(15,c)$ are parallel. Find c .

Big idea: Components must scale by the same factor.

Explanation

- 15 is 3 times 5, so c must be 3 times -2 .

Answer: $c=-6$

Formula sheet: Page 2 → vectors.

Common mistake: Use the same multiplier for every component.

T6 Q5

Question: If $a\cdot b=3$ and $a\times b=\langle -1, 0, \sqrt{8} \rangle$, find $\tan\theta$.

Big idea: $\tan\theta=\|a\times b\|/(a\cdot b)$.

Explanation

- $\|a\times b\|=\sqrt{1+0+8}=3$.

$$\tan\theta=3/3$$

Answer: 1

Formula sheet: Page 2 → Dot product and Cross product.

Common mistake: Use the magnitude of $a\times b$.

T6 Q6a

Question: Find vector of magnitude 7 perpendicular to $u=\langle 3, -2, 1 \rangle$ and $v=\langle 1, -2, 2 \rangle$.

Big idea: Cross product gives perpendicular direction; scale it.

Explanation

- $u\times v=\langle -2, -5, -4 \rangle$ and its magnitude is $3\sqrt{5}$.

- Multiply by $7/(3\sqrt{5})$.

Answer: $7(-2, -5, -4)/(3\sqrt{5})$, or its negative

Formula sheet: Page 2 → Cross product.

Common mistake: Cross product gives direction, not necessarily length 7.

T6 Q6b

Question: Write $\mathbf{b} = (1, 2, 3)$ as parallel plus orthogonal to $\mathbf{a} = (3, 6, -2)$.

Big idea: Use projection.

Explanation

$$\text{proj}_{\mathbf{a}}\mathbf{b} = (\mathbf{b} \cdot \mathbf{a} / \mathbf{a} \cdot \mathbf{a}) \mathbf{a} = 9/49(3, 6, -2) = (27/49, 54/49, -18/49)$$

- Orthogonal part is $\mathbf{b} - \text{proj}_{\mathbf{a}}\mathbf{b}$.

Answer: $(27/49, 54/49, -18/49) + (22/49, 44/49, 165/49)$

Formula sheet: Page 2 → Dot product.

Common mistake: Projection is the parallel part.

Tutorial 7 — Lines and planes

T7 Q1a

Question: Plane through $P(4, 3, 6)$, $Q(-2, 0, 8)$, $R(1, 5, 0)$.

Big idea: Two plane vectors give a normal by cross product.

Explanation

- $\mathbf{PQ} = (-6, -3, 2)$, $\mathbf{PR} = (-3, 2, -6)$.
- $\mathbf{PQ} \times \mathbf{PR} = (14, -42, -21)$, simplify to $(2, -6, -3)$.

$$2(x-4) - 6(y-3) - 3(z-6) = 0$$

Answer: $2x - 6y - 3z + 28 = 0$

Formula sheet: Page 2 → Cross product; Page 3 → Planes.

Common mistake: A plane needs a normal vector.

T7 Q1b

Question: Plane through $(1, 2, -3)$ perpendicular to line $x = 1 + 2t$, $y = 2 + t$, $z = -3 - 5t$.

Big idea: The line direction is the plane normal.

Explanation

- Line direction is $(2, 1, -5)$.

$$2(x-1) + (y-2) - 5(z+3) = 0$$

Answer: $2x + y - 5z - 19 = 0$

Formula sheet: Page 3 → Lines and Planes.

Common mistake: The plane normal comes from the line direction.

T7 Q2

Question: Line through $P(1, -1, 5)$, $Q(0, 1, 3)$.

Big idea: Direction is $\mathbf{Q} - \mathbf{P}$.

Explanation

- $\mathbf{Q} - \mathbf{P} = (-1, 2, -2)$.

Answer: $x = 1 - t$, $y = -1 + 2t$, $z = 5 - 2t$

Formula sheet: Page 3 → Lines.

Common mistake: Many parameter choices describe the same line.

T7 Q3

Question: Intersection line of $4x + 4y - 2z = 9$ and $2x + y + z = -3$.

Big idea: Direction is cross product of normals; find one point.

Explanation

- $n_1 = \langle 4, 4, -2 \rangle$ and $n_2 = \langle 2, 1, 1 \rangle$, so $n_1 \times n_2 = \langle 6, -8, -4 \rangle$.
- Set $z=0$ and solve the two equations to get $(-21/4, 15/2, 0)$.

Answer: $x = -21/4 + 6t$, $y = 15/2 - 8t$, $z = -4t$

Formula sheet: Page 2 → Cross product; Page 3 → Lines/Planes.

Common mistake: Need both a point and a direction.

T7 Q4

Question: Intersect line $x=1-3t$, $y=1-t$, $z=-1+2t$ with plane $-x+3y+3z=1$.

Big idea: Substitute the line into the plane.

Explanation

$$-(1-3t) + 3(1-t) + 3(-1+2t) = 1$$

- This gives $t = 1/3$. Substitute into the line.

Answer: $(0, 2/3, -1/3)$

Formula sheet: Page 3 → Lines and Planes.

Common mistake: Solve for t before giving the point.

T7 Q5a

Question: Line through $(2, 4, 1)$ perpendicular to plane $3x - y + 5z = 77$.

Big idea: Plane normal is line direction.

Explanation

- Normal vector is $\langle 3, -1, 5 \rangle$.

Answer: $x = 2 + 3t$, $y = 4 - t$, $z = 1 + 5t$

Formula sheet: Page 3 → Lines/Planes.

Common mistake: The constant 77 does not affect the normal.

T7 Q5b

Question: Angle between planes $x - 2y + z = 5$ and $2x + y - z = 0$.

Big idea: Use angle between normals.

Explanation

- $n_1 = \langle 1, -2, 1 \rangle$, $n_2 = \langle 2, 1, -1 \rangle$.

$$\cos \theta = |n_1 \cdot n_2| / (\|n_1\| \|n_2\|) = 1/6$$

Answer: $\theta = \arccos(1/6)$

Formula sheet: Page 2 → Dot product; Page 3 → Planes.

Common mistake: Use absolute value for the acute angle between planes.

T7 Q6a

Question: Distance from $A(1, -1, 3)$ to plane $-2x + y - 2z = 1$.

Big idea: Use point-plane distance.

Explanation

- Rewrite as $-2x + y - 2z - 1 = 0$.

$$\text{distance} = |-2(1) + (-1) - 2(3) - 1| / \sqrt{4 + 1 + 4} = 10/3$$

Answer: $10/3$

Formula sheet: Page 3 → Distances.

Common mistake: Use absolute value.

T7 Q6b

Question: Distance between $z = x + 2y + 1$ and $3x + 6y - 3z = 4$.

Big idea: Rewrite with same normal.

Explanation

- First plane: $x + 2y - z + 1 = 0$. Second: $x + 2y - z - 4/3 = 0$.

$$\text{distance} = |1 - (-4/3)| / \sqrt{1 + 4 + 1}$$

Answer: $7/(3\sqrt{6})$

Formula sheet: Page 3 → Distances.

Common mistake: Make normals identical before comparing constants.

T7 Q6c

Question: Distance from $Q(3, -1, 4)$ to line $x = -2 + 3t$, $y = -2t$, $z = 1 + 4t$.

Big idea: Use point-line distance formula.

Explanation

- Point on line $P = (-2, 0, 1)$, direction $v = \langle 3, -2, 4 \rangle$.

$$\text{distance} = \|(Q - P) \times v\| / \|v\|$$

Answer: $\sqrt{6}$

Formula sheet: Page 3 → Distances.

Common mistake: Use a point on the line, not the external point.

Tutorial 8 — Linear algebra

T8 Q1

Question: Solve $x_1 + x_2 - 2x_3 = 9$, $2x_1 + 4x_2 - 3x_3 = 1$, $3x_1 + 6x_2 - 5x_3 = 0$.

Big idea: Gaussian elimination gives a unique solution.

Explanation

- Write the augmented matrix and eliminate below pivots. Row reduction gives one pivot for each variable.

Answer: $x_1 = 25$, $x_2 = -10$, $x_3 = 3$

Formula sheet: Linear algebra row reduction.

Common mistake: Keep variable columns in order.

T8 Q2

Question: Show the 4-variable system has infinitely many solutions and find the solution set.

Big idea: One variable is free after row reduction.

Explanation

- RREF gives $x_1 + 2x_4 = 3$, $x_2 - 3x_4 = -5$, $x_3 + 4x_4 = 7$. Let $x_4 = t$.

Answer: $(x_1, x_2, x_3, x_4) = (3 - 2t, -5 + 3t, 7 - 4t, t)$

Formula sheet: Linear algebra row reduction.

Common mistake: A free variable means infinitely many solutions.

T8 Q3a1

Question: Compute $A + B^T$.

Big idea: Transpose B so dimensions match A.

Explanation

- B^T is 3×2 , same as A, so add entries.

Answer: $[[0, 4], [3, -1], [8, -8]]$

Formula sheet: Matrix arithmetic.

Common mistake: $A + B$ is not possible, but $A + B^T$ is.

T8 Q3a2

Question: Compute AC .

Big idea: Rows of A times columns of C.

Explanation

- A is 3×2 and C is 2×2 , so AC is 3×2 .

Answer: $[[2, 0], [7, 1], [21, 3]]$

Formula sheet: Matrix multiplication.

Common mistake: Matrix multiplication is not entry-by-entry.

T8 Q3b

Question: Find X if $\frac{3}{2}X + C = \begin{bmatrix} 3 & -4 \\ 5 & 4 \end{bmatrix}$.

Big idea: Solve like algebra with matrices.

Explanation

- Subtract C , then multiply by $\frac{2}{3}$.

$$X = \frac{2}{3} \begin{bmatrix} 1 & -4 \\ 8 & 5 \end{bmatrix}$$

Answer: $\begin{bmatrix} \frac{2}{3} & -\frac{8}{3} \\ 16/3 & 10/3 \end{bmatrix}$

Formula sheet: Matrix arithmetic.

Common mistake: Scalar multiplication applies to every entry.

T8 Q4

Question: Find h so A^{-1} does not exist.

Big idea: No inverse means determinant zero.

Explanation

- For the given matrix, $\det(A) = 12(h+6)$. Set this equal to zero.

Answer: $h = -6$

Formula sheet: Determinants/inverse criterion.

Common mistake: Do not set individual entries to zero.

T8 Q5

Question: Use cofactor expansion to find determinant.

Big idea: Expand along a row or column with zeros.

Explanation

- The second column has many zeros, so the expansion reduces to one 3×3 determinant.

Answer: 36

Formula sheet: Cofactor expansion.

Common mistake: Include the cofactor sign.

T8 Q6

Question: Given $\det(A) = 9$ and $\det(B) = 4$, evaluate determinant expressions.

Big idea: Use determinant properties.

Explanation

- $\det(kA) = k^3 \det(A)$ for 3×3 matrices. $\det(BA^{-1}) = \det(B) \det(A^{-1})$. Row scaling scales determinant; row replacement does not change determinant.

Answer: $\det(-3A) = -243$, $\det(BA^{-1}) = 4/9$, third determinant = 18, fourth determinant = -18

Formula sheet: Determinant properties.

Common mistake: For 3×3 , $\det(kA)$ uses k^3 .

Quiz 2 — Exam-style walkthrough

Quiz Q1a

Question: Evaluate $\int (x^4 + x^2 - 2)x^3 dx$.

Big idea: Divide first, then integrate.

Explanation

$$(x^4 + x^2 - 2)x^3 = x^7 + x^5 - 2x^3$$

$$\int = \frac{x^8}{8} + \frac{x^6}{6} - \frac{2x^4}{4} + C$$

Answer: $\frac{x^8}{8} + \frac{x^6}{6} - \frac{1}{2}x^4 + C$

Formula sheet: Page 1 → Basic integration.

Common mistake: The $1/x$ term gives $\ln|x|$.

Quiz Q1b(i)

Question: Evaluate $\int_0^3 x\sqrt{x^2+1}dx$.

Big idea: Substitution with bounds.

Explanation

- Let $u=x^2+1$, $du=2xdx$. Bounds $0 \rightarrow 1$ and $3 \rightarrow 10$.

$$\frac{1}{2} \int_1^{10} u^{1/2} du = \frac{1}{3} (10^{3/2} - 1)$$

Answer: $\frac{1}{3}(10^{3/2}-1)$

Formula sheet: [Page 1](#) $\rightarrow$ Basic integration.

Common mistake: Change the bounds.

Quiz Q1b(ii)

Question: Evaluate $\int \sin^3 x \cos^3 x \, dx$.

Big idea: Odd powers allow substitution.

Explanation

- Write $\sin^3 x = (1 - \cos^2 x) \sin x$ and let $u = \cos x$.

$$\int (u^5 - u^3) du$$

Answer: $\cos^6 x / 6 - \cos^4 x / 4 + C$

Formula sheet: [Page 1](#) $\rightarrow$ Pythagorean identities.

Common mistake: Track the negative sign from du .

Quiz Q2

Question: $F(x) = \int_2^{3x^2} \sqrt{t^2+25} dt$. Find $F'(2)$.

Big idea: FTC with upper limit.

Explanation

$$F'(x) = 6x\sqrt{(3x^2)^2+25} = 6x\sqrt{9x^4+25}$$

$$F'(2) = 12\sqrt{169} = 156$$

Answer: 156

Formula sheet: [Page 1](#) $\rightarrow$ FTC.

Common mistake: Do not integrate the radical.

Quiz Q3a

Question: Find x, y if $(2+i)x + (1-2i)y = 8-i$.

Big idea: Equate real and imaginary parts.

Explanation

$$(2x+y) + i(x-2y) = 8-i$$

- So $2x+y=8$ and $x-2y=-1$.

Answer: $x=3$, $y=2$

Formula sheet: [Page 2](#) $\rightarrow$ Complex numbers.

Common mistake: Imaginary equation comes from coefficient of i .

Quiz Q3b

Question: Find $\sqrt{7+24i}$.

Big idea: Let the square root be $a+bi$.

Explanation

- $(a+bi)^2 = a^2 - b^2 + 2abi$, so $a^2 - b^2 = 7$ and $2ab = 24$.

- $a=4$, $b=3$ works.

Answer: $4+3i$

Formula sheet: [Page 2](#) $\rightarrow$ Complex numbers.

Common mistake: There are two roots; this is the main one.

Quiz Q4

Question: Sphere with diameter $A(-4, -1, 2)$, $B(2, 3, 6)$.

Big idea: Midpoint and radius.

Explanation

- Center is $(-1, 1, 4)$. $AB = \sqrt{68}$, so $r^2 = 17$.

Answer: $(x+1)^2 + (y-1)^2 + (z-4)^2 = 17$

Formula sheet: Page 2 → 3D space.

Common mistake: Right side is r^2 .

Quiz Q5

Question: Unit vector perpendicular to $u = \langle 1, 1, 1 \rangle$ and $v = \langle 1, -1, 0 \rangle$.

Big idea: Cross product then normalize.

Explanation

- $u \times v = \langle 1, 1, -2 \rangle$, magnitude $\sqrt{6}$.

Answer: $\langle 1/\sqrt{6}, 1/\sqrt{6}, -2/\sqrt{6} \rangle$, or its negative

Formula sheet: Page 2 → Cross product.

Common mistake: Perpendicular does not automatically mean unit length.

Quiz Q6

Question: $\|a\| = 3$, $\|b\| = 2$, $\theta = 2\pi/3$. Find $\|a+b\|$.

Big idea: Use vector-sum magnitude.

Explanation

$$\|a+b\|^2 = 3^2 + 2(3)(2)\cos(2\pi/3) + 2^2 = 9 - 6 + 4 = 7$$

Answer: $\sqrt{7}$

Formula sheet: Page 2 → Dot product.

Common mistake: Use $\cos(2\pi/3) = -1/2$.

Final exam practice questions

Practice Q1.1

Question: Evaluate $\int (x\sqrt{x} + 3x - 1) dx$.

Big idea: Rewrite $x\sqrt{x} = x^{3/2}$.

Explanation

- Rewrite $x\sqrt{x} = x^{3/2}$.

Answer: $2/5 x^{5/2} + 3/2 x^2 - x + C$

Formula sheet: Page 1 → Basic integration.

Common mistake: Write enough method steps to earn partial credit.

Practice Q1.2

Question: Evaluate $\int_1^2 (2/x - 3/x^2) dx$.

Big idea: Antiderivative is $2\ln|x| + 3/x$, then evaluate 1 to 2.

Explanation

- Antiderivative is $2\ln|x| + 3/x$, then evaluate 1 to 2.

Answer: $2\ln 2 - 3/2$

Formula sheet: Page 1 → Basic integration.

Common mistake: Write enough method steps to earn partial credit.

Practice Q2.1

Question: Evaluate $\int_0^1 x^3 \sqrt{x^4 + 1} dx$.

Big idea: Use $u = x^4 + 1$, $du = 4x^3 dx$, bounds 1 to 2.

Explanation

- Use $u = x^4 + 1$, $du = 4x^3 dx$, bounds 1 to 2.

Answer: $1/6(2^{3/2} - 1)$

Formula sheet: Page 1 → substitution.

Common mistake: Write enough method steps to earn partial credit.

Practice Q2.2

Question: Evaluate $\int \sin^3 x \sqrt{\cos x} dx$.

Big idea: Use $\sin^3 x = (1 - \cos^2 x) \sin x$ and $u = \cos x$.

Explanation

- Use $\sin^3 x = (1 - \cos^2 x) \sin x$ and $u = \cos x$.

Answer: $\frac{2}{7} \cos^{7/2} x - \frac{2}{3} \cos^{3/2} x + C$

Formula sheet: Page 1 → Pythagorean identities.

Common mistake: Write enough method steps to earn partial credit.

Practice Q3

Question: $F(x) = \int$ from x^2 to $5x$ of $1/\sqrt{t+1} dt$. Find $F'(1)$.

Big idea: Use upper minus lower FTC: $F'(x) = 5/\sqrt{5x+1} - 2x/\sqrt{x^2+1}$.

Explanation

- Use upper minus lower FTC: $F'(x) = 5/\sqrt{5x+1} - 2x/\sqrt{x^2+1}$.

Answer: $5\sqrt{6} - \sqrt{2}$

Formula sheet: Page 1 → FTC.

Common mistake: Write enough method steps to earn partial credit.

Practice Q4.1

Question: Find x, y if $(x+iy)(2+i) = 3-i$.

Big idea: Expand to $(2x-y) + i(x+2y)$, then match parts.

Explanation

- Expand to $(2x-y) + i(x+2y)$, then match parts.

Answer: $x=1, y=-1$

Formula sheet: Page 2 → Complex numbers.

Common mistake: Write enough method steps to earn partial credit.

Practice Q4.2

Question: Find x, y if $x+2iy-8 = i(7-2x+2iy)$.

Big idea: Right side is $-2y+i(7-2x)$, then match parts.

Explanation

- Right side is $-2y+i(7-2x)$, then match parts.

Answer: $x=-1, y=9/2$

Formula sheet: Page 2 → Complex numbers.

Common mistake: Write enough method steps to earn partial credit.

Practice Q5

Question: Find $\sqrt{-5-12i}$.

Big idea: Let $a+bi$; solve $a^2-b^2=-5$ and $2ab=-12$.

Explanation

- Let $a+bi$; solve $a^2-b^2=-5$ and $2ab=-12$.

Answer: $-2+3i$

Formula sheet: Page 2 → Complex numbers.

Common mistake: Write enough method steps to earn partial credit.

Practice Q6

Question: Show $x^2+y^2+z^2=4x-6y+2z+1$ is a sphere.

Big idea: Complete squares: $(x-2)^2+(y+3)^2+(z-1)^2=15$.

Explanation

- Complete squares: $(x-2)^2+(y+3)^2+(z-1)^2=15$.

Answer: center $(2,-3,1)$, radius $\sqrt{15}$

Formula sheet: Page 2 → 3D space.

Common mistake: Write enough method steps to earn partial credit.

Practice Q7

Question: Sphere with diameter $A(-1,2,3)$, $B(2,0,5)$.

Big idea: Center is $(\frac{1}{2},1,4)$; diameter length $\sqrt{17}$, so $r^2=17/4$.

Explanation

- Center is $(\frac{1}{2},1,4)$; diameter length $\sqrt{17}$, so $r^2=17/4$.

Answer: $(x-\frac{1}{2})^2+(y-1)^2+(z-4)^2=17/4$

Formula sheet: Page 2 → 3D space.

Common mistake: Write enough method steps to earn partial credit.

Practice Q8

Question: $\|a\|=2$, $\|b\|=\sqrt{2}$, $\theta=\pi/4$. Find $\|a+b\|$.

Big idea: $\|a+b\|^2=4+2+2(2)(\sqrt{2})\cos(\pi/4)=10$.

Explanation

- $\|a+b\|^2=4+2+2(2)(\sqrt{2})\cos(\pi/4)=10$.

Answer: $\sqrt{10}$

Formula sheet: Page 2 → Dot product.

Common mistake: Write enough method steps to earn partial credit.

Practice Q9.1

Question: Find x for perpendicular vectors $\langle 2x-1, 3x, \frac{1}{2} \rangle$ and $\langle 3, -2, 2x \rangle$.

Big idea: Set dot product to zero: $3(2x-1)-2(3x)+\frac{1}{2}(2x)=0$.

Explanation

- Set dot product to zero: $3(2x-1)-2(3x)+\frac{1}{2}(2x)=0$.

Answer: $x=3$

Formula sheet: Page 2 → Dot product.

Common mistake: Write enough method steps to earn partial credit.

Practice Q9.2

Question: Area of triangle $P(1,3,5)$, $Q(-2,1,2)$, $R(1,1,1)$.

Big idea: $PQ=\langle -3, -2, -3 \rangle$, $PR=\langle 0, -2, -4 \rangle$, $PQ \times PR=\langle 2, -12, 6 \rangle$.

Explanation

- $PQ=\langle -3, -2, -3 \rangle$, $PR=\langle 0, -2, -4 \rangle$, $PQ \times PR=\langle 2, -12, 6 \rangle$.

Answer: $\sqrt{46}$

Formula sheet: Page 2 → Cross product.

Common mistake: Write enough method steps to earn partial credit.

Practice Q10

Question: Line of intersection of $4x+4y-2z=9$ and $2x+y+z=-3$.

Big idea: Direction is $\langle 6, -8, -4 \rangle$; one point is $(-21/4, 15/2, 0)$.

Explanation

- Direction is $\langle 6, -8, -4 \rangle$; one point is $(-21/4, 15/2, 0)$.

Answer: $x=-21/4+6t$, $y=15/2-8t$, $z=-4t$

Formula sheet: Page 3 → Lines/Planes.

Common mistake: Write enough method steps to earn partial credit.

Practice Q11a

Question: Plane through $A(2,0,-1)$, $B(1,-1,1)$, $C(0,3,-2)$.

Big idea: $AB \times AC = \{-5, -5, -5\}$, so use normal $\{1, 1, 1\}$.

Explanation

- $AB \times AC = \{-5, -5, -5\}$, so use normal $\{1, 1, 1\}$.

Answer: $x+y+z=1$

Formula sheet: Page 3 → Planes.

Common mistake: Write enough method steps to earn partial credit.

Practice Q11b

Question: Distance from $D(5, -2, -1)$ to $x+y+z=1$.

Big idea: Use $|x_0+y_0+z_0-1|/\sqrt{3}$.

Explanation

- Use $|x_0+y_0+z_0-1|/\sqrt{3}$.

Answer: $1/\sqrt{3}$

Formula sheet: Page 3 → Distances.

Common mistake: Write enough method steps to earn partial credit.

Practice Q12

Question: Solve $3x-y-5z=3$, $4x-4y-3z=-4$, $x-5z=2$.

Big idea: Row reduction gives a unique solution.

Explanation

- Row reduction gives a unique solution.

Answer: $x=2$, $y=3$, $z=0$

Formula sheet: Linear algebra.

Common mistake: Write enough method steps to earn partial credit.

Practice Q13

Question: Solve truss system $2x_1-2x_2+x_3=3$, $3x_1+x_2-x_3=7$, $x_1-3x_2+2x_3=0$.

Big idea: Row reduction gives the force variables.

Explanation

- Row reduction gives the force variables.

Answer: $x_1=2$, $x_2=0$, $x_3=-1$

Formula sheet: Linear algebra.

Common mistake: Write enough method steps to earn partial credit.

Practice Q14

Question: Find determinant of the 4×4 matrix.

Big idea: Expand along the row or column with most zeros and include cofactor signs.

Explanation

- Expand along the row or column with most zeros and include cofactor signs.

Answer: -6

Formula sheet: Determinants.

Common mistake: Write enough method steps to earn partial credit.

Practice Q15a1

Question: Compute $A+B^T$.

Big idea: Transpose B , then add matching entries.

Explanation

- Transpose B , then add matching entries.

Answer: $[[0, 4], [3, -1], [8, -8]]$

Formula sheet: Matrix arithmetic.

Common mistake: Write enough method steps to earn partial credit.

Practice Q15a2

Question: Compute AC.

Big idea: Rows of A times columns of C.

Explanation

- Rows of A times columns of C.

Answer: $\begin{bmatrix} 2 & 0 \\ 7 & 1 \\ 21 & 3 \end{bmatrix}$

Formula sheet: Matrix multiplication.

Common mistake: Write enough method steps to earn partial credit.

Practice Q15b

Question: Find X if $\frac{2}{3}X + C = \begin{bmatrix} 3 & -4 \\ 5 & 4 \end{bmatrix}$.

Big idea: Subtract C, then multiply by $\frac{3}{2}$.

Explanation

- Subtract C, then multiply by $\frac{3}{2}$.

Answer: $\begin{bmatrix} \frac{1}{3} & -\frac{8}{3} \\ 16/3 & 10/3 \end{bmatrix}$

Formula sheet: Matrix arithmetic.

Common mistake: Write enough method steps to earn partial credit.

Final checklist

- For integration, rewrite radicals and fractions first.
- For substitution, state u and du; change bounds for definite integrals.
- For trig integrals, save the factor that becomes du.
- For FTC, upper limit is positive and lower limit is subtracted.
- For distance travelled, integrate $|v(t)|$.
- For complex numbers, match real and imaginary parts.
- For spheres, complete the square.
- For planes, find a normal; for lines, find a direction.
- For dot product: angles and perpendicularity. For cross product: perpendicular vectors, areas, and plane normals.
- For matrices, check dimensions. For inverse existence, use $\det(A) \neq 0$ when asking when no inverse exists.